Notes: Bennedsen, Nielsen, Pérez-González, Wolfenzon (QJE, 2007)

Inside the Family Firm: The Role of Families in Succession Decisions and Performance

Contents

1 Big picture

1.1 Research question

Family firms are prevalent worldwide, and many are run by founders or their descendants. The paper asks:

> Do firms perform worse when they appoint a *family CEO* (a successor related to the departing CEO), and if so, is the effect causal?

The difficulty is endogeneity: families might choose heirs precisely when prospects are good (or bad), and family firms differ systematically from nonfamily firms.

1.2 Core contribution

The paper's contribution is to deliver **credible causal evidence** on the effect of family CEOs on firm performance by combining:

- **Unique data:** Danish registries link individuals to firms and to family members (children, spouses, etc.).
- **A quasi-experiment:** the gender of the departing CEO's firstborn child predicts family succession decisions but is plausibly unrelated to firm performance determinants.
- **A CEO-turnover design:** performance is measured *around transitions*, reducing sensitivity to time-invariant firm heterogeneity.

1.3 One-line finding

Instrumenting family CEO appointments with firstborn gender, the paper finds that appointing a family CEO causes a **large drop in operating profitability** around CEO transitions.

2 Data, sample, and definitions

2.1 Sample of CEO transitions

The core dataset is a panel of Danish firms with observed CEO transitions. The paper constructs a transition-level dataset indexed by i (a firm succession event).

Key sample sizes (from the tables):

- Total CEO successions: 5,334 (Table II).
- Successions with nonmissing firstborn gender (departing CEO has at least one child and the gender is observed): 4,692 (Tables III–VI).

2.2 Family CEO definition

Define:

$$famCEO_i \equiv \mathbf{1}\{\text{incoming CEO is related by blood or marriage to the departing CEO}\}.$$

Family successions include:

- **Family: children** (successor is a child of the departing CEO),
- **Family: others** (successor is another relative by blood or marriage).

2.3 Performance measures

The main performance measures are operating profitability ratios computed from accounting data:

- **OROA:** operating income divided by book assets.
- **Industry-adjusted OROA:** $OROA - \overline{OROA}^{industry}$ (four-digit NACE benchmark).
- **Industry-and-performance-adjusted OROA:** Barber–Lyon style benchmark adjusting for both industry and pre-transition performance.

To focus on CEO-induced changes, the key dependent variable is the change in profitability around the transition:

$$y_i \equiv \overline{Perf}_i^{post} - \overline{Perf}_i^{pre},$$

where the pre and post windows are typically three-year averages excluding the transition year.

3 Empirical strategy and models

3.1 Model 1: baseline turnover regression (Equation (1))

The starting point is a regression of performance changes on whether the successor is a family CEO:

$$y_i \;=\; a_1 + X_i' b_1 + c_1\, famCEO_i + \varepsilon_{1i}. \tag{1}$$

Interpretation. c_1 is the difference in the pre-to-post profitability change between family successions and unrelated successions, controlling for X_i.

Why this is not necessarily causal. Even though y_i differences out time-invariant firm heterogeneity, the decision to appoint a family CEO may be correlated with time-varying prospects, mean reversion, or unobserved succession-specific shocks.

3.2 Instrumental variables design

The paper uses **the gender of the departing CEO's firstborn child** as an instrument for $famCEO_i$.

Define the instrument:

$$Z_i \equiv genderfirst_i \equiv \mathbf{1}\{\text{firstborn child of departing CEO is male}\}.$$

3.2.1 Relevance intuition

If families have a preference for male heirs, a male firstborn increases the probability of selecting a family CEO, especially a child successor. Table II and Table V provide this evidence.

3.2.2 Exclusion intuition

Firstborn gender is determined at birth and (locally) should not predict firm performance changes around a later CEO transition, except through its effect on succession choices. Table III provides balance tests supporting this.

3.2.3 Model 2: first stage (Equation (2))

$$famCEO_i \;=\; a_2 + X_i' b_2 + c_2\, Z_i + \varepsilon_{2i}. \tag{2}$$

The first stage is estimated using OLS (linear probability model), following standard IV practice for binary treatments.

3.2.4 Model 3: second stage (Equation (3))

$$y_i \;=\; a_3 + X_i' b_3 + c_3\, \widehat{famCEO}_i + \varepsilon_{3i}. \tag{3}$$

Interpretation. c_3 is the causal effect of family CEOs on performance changes for **compliers** (firms whose CEO choice is shifted by firstborn gender).

3.2.5 LATE interpretation (derivation)

Under standard IV assumptions (relevance, independence, exclusion, monotonicity),

$$c_3 \;=\; \frac{\mathbb{E}[y_i \mid Z_i = 1] - \mathbb{E}[y_i \mid Z_i = 0]}{\mathbb{E}[famCEO_i \mid Z_i = 1] - \mathbb{E}[famCEO_i \mid Z_i = 0]} \;=\; \frac{\text{Reduced Form}}{\text{First Stage}}. \tag{4}$$

Table V directly reports both objects: the first-stage jump in $famCEO$ from a male firstborn and the reduced-form effect on profitability changes.

3.3 Alternative sources of identification used for robustness

Beyond the firstborn gender instrument, the paper uses:

- alternative male-child instruments (any son, number of sons, male share of children; Table V and Table VI),
- a **CEO death around transition** instrument, which creates an arguably exogenous turnover (Table VII).

TABLE I
FIRM CHARACTERISTICS BY TYPE OF CEO SUCCESSION

		Type of Succession		
Variable	All (1)	Family (2)	Unrelated (3)	Difference (4)
Ln assets	8.605	8.232	8.791	−0.559***
	(0.0240)	(0.0332)	(0.0315)	(0.0458)
	[5,334]	[1,776]	[3,558]	
Operating return on assets (OROA)	0.065	0.074	0.061	0.013***
	(0.0020)	(0.0032)	(0.0025)	(0.0041)
	[5,334]	[1,776]	[3,558]	
Net income to assets	0.033	0.038	0.031	0.007*
	(0.0019)	(0.0031)	(0.0024)	(0.0039)
	[5,334]	[1,776]	[3,558]	
Industry-adjusted OROA	−0.002	0.007	−0.006	0.014***
	(0.0020)	(0.0032)	(0.0025)	(0.0041)
	[5,334]	[1,776]	[3,558]	
Firm Age	19.417	19.826	19.213	0.613
	(0.3106)	(0.4840)	(0.3981)	(0.6267)
	[5,334]	[1,776]	[3,558]	

4 Interpreting all tables (I–X)

4.1 Table I: firm characteristics by type of CEO succession

What it reports. Means at the time of transition for all firms and split by whether the successor is family or unrelated.

Key patterns.

- **Size:** family-succession firms are smaller (mean ln(assets) is 8.232 vs 8.791; difference −0.559***).
- **Pre-transition profitability:** family-succession firms are *more* profitable at transition (OROA is 0.074 vs 0.061; difference 0.013***), and the same holds after industry adjustment.

Interpretation. Selection is nontrivial: families tend to keep control in relatively smaller but *better-performing* firms. This selection is exactly why OLS is likely to understate the causal cost of family CEOs.

4.2 Table II: successions and departing CEO family structure

What it reports. Counts and shares of family vs unrelated successions by the departing CEO's family structure (spouses, children, gender composition of children, firstborn gender).

Key messages.

- **Children matter.** More children increases the likelihood of family succession, especially child successions.
- **Male children matter.** Having a higher male share of children is associated with more family successions.
- **Firstborn gender is powerful.** A male firstborn increases the family-succession rate by about 9.6 percentage points (Panel D), and the effect is concentrated in successions to children (about 10.8 percentage points).
- **Family instability reduces family succession.** Multiple spouses are associated with fewer family successions (Panel A).

TABLE II
FIRM SUCCESSIONS AND FAMILY CHARACTERISTICS OF DEPARTING CEOS

		Type of succession							
		Family		Unrelated		Family: Children		Family: Others	
Description	Number of successions	Number	Share	Number	Share	Number	Share	Number	Share
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
All	5,334	1,776	0.333	3,558	0.667	863	0.162	913	0.171
A. Number of spouses:									
0	434	79	0.182	355	0.618	5	0.012	74	0.171
1	4,282	1,541	0.360	2,741	0.640	602	0.167	739	0.173
2 or more	618	156	0.252	462	0.748	56	0.091	100	0.162
Difference (2 or more) minus (1)			**−0.107*****				**−0.097*****		**−0.011**
			(0.020)				(0.013)		(0.016)
B. Number of children:									
0	642	159	0.248	483	0.752	—	0.000	159	0.248
1	807	235	0.291	572	0.709	96	0.119	139	0.172
2	2,397	770	0.321	1,627	0.679	389	0.162	381	0.159
3	1,152	476	0.413	676	0.587	296	0.257	180	0.156
4 or more	336	136	0.405	200	0.595	82	0.244	54	0.161
Difference (1) minus (0)			**0.044***				**0.119*****		−0.075***
			(0.023)				(0.011)		(0.022)
Difference (3) minus (1)			**0.122*****				**0.138*****		−0.016
			(0.022)				(0.017)		(0.017)
C. By gender ratio (male/children):									
<50 percent	1,511	437	0.289	1,074	0.711	161	0.107	276	0.183
=50 percent	1,345	451	0.335	894	0.665	248	0.184	203	0.151
>50 percent	1,836	729	0.397	1,107	0.603	454	0.247	275	0.150
Difference (>50 percent) minus (<50 percent)			**0.108*****				**0.141*****		−0.033**
			(0.016)				(0.013)		(0.013)
D. By gender of first born child									
Female	2,216	652	0.294	1,564	0.706	281	0.127	371	0.167
Male	2,476	965	0.390	1,511	0.610	582	0.235	383	0.155
Difference male minus female			**0.096*****				**0.108*****		−0.013
			(0.014)				(0.011)		(0.011)

Interpretation. Table II establishes instrument relevance and supports the "male heir" channel: firstborn gender is a strong predictor of whether a family member becomes CEO.

4.3 Table III: balance tests by firstborn gender

What it reports. Firm and family characteristics split by whether the departing CEO's first child is male or female.

Main conclusion. Most firm and family observables are statistically similar across $Z_i = 1$ and $Z_i = 0$:

- firm size, OROA, net income/assets, and firm age are not meaningfully different across groups,
- the number of children is essentially identical.

Interpretation. This supports the plausibility of the exclusion restriction: firstborn gender appears orthogonal to observable determinants of performance near the transition.

4.4 Table IV: performance around CEO transitions (DD evidence)

Part A. Reports average industry-adjusted OROA in the three years before and after transitions. Unrelated successions show a strong improvement after transition (about $+0.0132^{***}$), while family successions show essentially no improvement (about -0.0010). **Part B.** Computes the difference-in-differences across succession types. Across multiple outcomes, family successions underperform:

- raw OROA DD: about -0.0154^{***},
- industry-adjusted OROA DD: about -0.0141^{***},

TABLE III
FIRM AND FAMILY CHARACTERISTICS BY THE GENDER OF THE FIRST CHILD OF DEPARTING CEOS

Variable	All (1)	Gender of First Child: Male (2)	Gender of First Child: Female (3)	Difference (4)
Ln assets	8.638	8.617	8.662	−0.045
	(0.0255)	(0.0352)	(0.0369)	(0.0510)
	[4,692]	[2,476]	[2,216]	
Operating return on assets (OROA)	0.067	0.066	0.069	−0.003
	(0.0022)	(0.0030)	(0.0031)	(0.0043)
	[4,692]	[2,476]	[2,216]	
Net income to assets	0.035	0.033	0.037	−0.004
	(0.0020)	(0.0028)	(0.0029)	(0.0040)
	[4,692]	[2,476]	[2,216]	
Industry-adjusted OROA	−0.0003	−0.0028	0.0024	−0.0052
	(0.0021)	(0.0030)	(0.0031)	(0.0043)
	[4,692]	[2,476]	[2,216]	
Firm age	19.247	19.307	19.180	0.127
	(0.3175)	(0.4370)	(0.4621)	(0.6361)
	[4,692]	[2,476]	[2,216]	
Number of children of departing CEO	2.236	2.240	2.231	0.009
	(0.0127)	(0.0175)	(0.0184)	(0.0253)
	[4,692]	[2,476]	[2,216]	
Departing CEO marital status is divorced	0.067	0.067	0.067	0.000
	(0.0037)	(0.0050)	(0.0053)	(0.0073)
	[4,692]	[2,476]	[2,216]	
Number of spouses of departing CEO	1.100	1.109	1.091	0.018
	(0.0063)	(0.0089)	(0.0087)	(0.0125)
	[4,692]	[2,476]	[2,216]	

TABLE IV
CEO SUCCESSION DECISIONS AND FIRM PERFORMANCE AROUND CEO TRANSITIONS

Part A. Dependent Variable: Industry-Adjusted Operating Return on Assets (OROA)

	All (1)	Type of succession: Family (2)	Type of succession: Unrelated (3)	Difference (4)
Before	−0.0032	0.0077	−0.0085	0.0162***
	(0.0016)	(0.0024)	(0.0020)	(0.0031)
	[5,334]	[1,776]	[3,558]	
After	0.0053	0.0067	0.0046	0.0021
	(0.0016)	(0.0026)	(0.0020)	(0.0033)
	[5,334]	[1,776]	[3,558]	
Difference	0.0084***	−0.0010	0.0132***	−0.0141***
	(0.0018)	(0.0028)	(0.0023)	(0.0036)

Part B. Alternative Dependent Variables (Difference-in-Differences (DD) analysis)

Differences in	Type of succession: Family (1)	Type of succession: Unrelated (2)	Mean Difference-in-differences (3)	Median DD (4)
Operating return on assets (OROA)	−0.0120***	0.0035	**−0.0154*** **	**−0.0082*** **
	(0.0028)	(0.0023)	**(0.0036)**	**(0.0027)**
Industry-adjusted OROA	−0.0010	0.0132***	**−0.0141*** **	**−0.0071*** **
	(0.0028)	(0.0023)	**(0.0036)**	**(0.0027)**
Industry-and-performance-adjusted OROA	0.0009	0.0107***	**−0.0098*** **	**−0.0066*** **
	(0.0027)	(0.0021)	**(0.0034)**	**(0.0025)**
Industry-adjusted net income to assets	−0.0056*	0.0064***	**−0.0120*** **	**−0.0060*** **
	(0.0029)	(0.0022)	**(0.0036)**	**(0.0023)**
Ln assets	0.0092***	0.0300***	**−0.0208*** **	**−0.0050*** **
	(0.0022)	(0.0019)	**(0.0029)**	**(0.0019)**
Ln sales	0.0003	0.0216***	**−0.0213*** **	**−0.0057** **
	(0.0059)	(0.0038)	**(0.0070)**	**(0.0025)**

TABLE V
GENDER OF THE FIRSTBORN CHILD, FAMILY SUCCESSIONS, AND PERFORMANCE

Part A. First Stage	Dependent variable: family CEO (1)	(2)	(3)	(4)	(5)	(6)	(7)
Gender of the first born child is male	0.0955***	0.0404**			0.0955***	0.0927***	0.0936***
	(0.0138)	(0.0171)			(0.0136)	(0.0135)	(0.0135)
Male child indicator variable		0.1162***					
		(0.0191)					
Number of male children			0.0737***				
			(0.0077)				
Ratio male to total children				0.1436***			
				(0.0186)			
Ln assets					−0.0448***	−0.0515***	−0.0508***
					(0.0034)	(0.0036)	(0.0037)
Firm age						0.0016***	0.0015***
						(0.0003)	(0.0003)
Industry-adjusted OROA, t = −1						0.2446***	
						(0.0445)	
Industry-and-performance-adjusted OROA, t = −1							0.3374***
							(0.0792)
Year controls	No	No	No	No	Yes	Yes	Yes
F-statistic	48.058	46.566	91.768	59.494	25.590	26.506	24.662
Number of CEO transitions	4,692	4,692	4,692	4,692	4,692	4,692	4,692

Part B. Reduced Form	Dependent variable: differences in operation-profitability around CEO successions (Three-year average postsuccession) − (three-year average pretransition) (1)	(2)	(3)	(4)	(5)	(6)	(7)
Gender of the first born child is male	−0.0120***	−0.0123***			−0.0121***	−0.0086**	−0.0083**
	(0.0038)	(0.0045)			(0.0038)	(0.0035)	(0.0034)
Male child indicator variable		0.0006					
		(0.0054)					
Number of male children			−0.0045**				
			(0.0022)				
Ratio male to total children				−0.0116**			
				(0.0053)			
Ln assets					−0.0040***	−0.0027***	−0.0029***
					(0.0010)	(0.0010)	(0.0010)
Firm age						−0.0000	0.0000
						(0.0001)	(0.0001)
Industry-adjusted OROA, t = −1						−0.3737***	
						(0.0163)	
Industry-and-performance-adjusted OROA, t = −1							−0.4219***
							(0.0311)
Year controls	No	No	No	No	Yes	Yes	Yes
Number of CEO transitions	4,692	4,692	4,692	4,692	4,692	4,692	4,692

- industry-and-performance-adjusted OROA DD: about -0.0098^{***},
- net income/assets DD: about -0.0120^{***},
- growth proxies (ln assets, ln sales) are also lower for family successions.

Interpretation. Even before IV, the turnover-based DD design suggests that unrelated CEOs are associated with post-transition performance improvements that family CEOs do not deliver.

4.5 Table V: instrument relevance (first stage) and reduced form

Part A: First stage. The firstborn-male indicator increases the probability of a family CEO by about 9–10 percentage points in the simplest specification, and remains large and highly significant with controls and year effects. F-statistics are large (well above common weak-instrument thresholds). Including "has any son" as a control reduces the incremental firstborn-male coefficient (consistent with the male-heir channel rather than purely "having a son").

Part B: Reduced form. A male firstborn predicts a *more negative* change in profitability around the transition (about -0.012 in baseline), consistent with the chain:

$$\text{male firstborn} \Rightarrow \text{more family CEOs} \Rightarrow \text{worse performance changes.}$$

Interpretation. Table V provides the two ingredients for IV: strong relevance (Part A) and a nonzero reduced form (Part B).

4.6 Table VI: main IV results (OLS vs IV-2SLS)

TABLE VI
FIRM PERFORMANCE AND FAMILY SUCCESSIONS: OLS AND INSTRUMENTAL VARIABLES

	Dependent variable: differences in operating profitability around CEO successions (Three-year average postsuccession) − (Three-year average pretransition)							
	OLS				IV-2SLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Family CEO	−0.0142***	−0.0079**	−0.1260***	−0.0722**	−0.0606**	−0.0808**	−0.0928**	−0.0886**
	(0.0038)	(0.0036)	(0.0429)	(0.0317)	(0.0298)	(0.0383)	(0.0393)	(0.0384)
Ln assets		−0.0030***					−0.0074***	−0.0074***
		(0.0010)					(0.0023)	(0.0022)
Firm age		0.0000					0.0001	0.0002
		(0.0001)					(0.0001)	(0.0001)
Industry-adjusted OROA, t = −1		−0.3727***					−0.3510***	
		(0.0164)					(0.0198)	
Industry-and-performance-adjusted OROA, t = −1								−0.3920***
								(0.0350)
Year controls	No	Yes	No	No	No	No	Yes	Yes
Number of CEO transitions	4,692	4,692	4,692	4,692	4,692	4,692	4,692	4,692
Instrumental variables								
Gender of the first child			✓	✓			✓	✓
Male child indicator variable				✓				
Number of male children					✓			
Ratio male to total children						✓		

TABLE VII
ALTERNATIVE WINDOWS OF ANALYSIS AND SUBSAMPLES BASED ON INDIVIDUAL CHARACTERISTICS

	Dependent variable: differences in industry-and-performance-adjusted operating profitability (Three-year average postsuccession) − (three-year average pretransition)						
	Windows			Sub-samples			
	Pretransition differences in performance	Differences in performance around CEO transitions	Posttransition differences in performance	Departing CEO's age (55, 70)	Departing CEO's age other than (55, 70)	Full sample	CEO transition and departing CEO death occur in the same year
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Family CEO	0.0325	−0.0886**	−0.0435	−0.0836**	−0.0962	−0.0372***	−0.1475
	(0.0420)	(0.0384)	(0.0608)	(0.0382)	(0.0926)	(0.0141)	(0.1411)
Year controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of CEO transitions	2,480	4,692	1,511	2,159	2,533	5,334	447
Instrumental variables							
Gender of the first child	✓	✓	✓	✓	✓		✓
Death of CEO around transition						✓	

What it reports. Estimates of the effect of $famCEO$ on the change in operating profitability around the transition.

OLS results. OLS suggests a negative but relatively small effect (about -0.014 without year FE; about -0.008 with year FE).

IV results. Instrumenting with firstborn gender yields a much larger negative effect (e.g. -0.126 in the simplest IV specification). Across alternative instruments and controls, the IV effects remain economically large (roughly -0.06 to -0.13). The preferred specification using industry-and-performance-adjusted OROA reports about -0.0886^{**}.

Interpretation. The substantial IV–OLS gap indicates strong selection: families tend to appoint family CEOs in situations where OLS comparisons would otherwise look less negative (or

TABLE VIII

OTHER SUBSAMPLES BASED ON FIRM AND INDUSTRY CHARACTERISTICS

	Dependent variable: differences in industry-and-performance-adjusted operating profitability (Three-year average postsuccession) − (Three-year average pretransition)										
		Firm characteristics			Industry characteristics						
	Full sample	Firms with assets ≥ median	Firms with formal board of directors	Firms with family members in board postsuccession	High share of family CEO transitions	High growth	Reports R&D spending	High per worker wages	High labor force schooling levels	High Import penetration	High output volatility
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Family CEO	$\boxed{-0.0886^{**}}$	$\boxed{-0.1140^{**}}$	$\boxed{-0.0934^{**}}$	$\boxed{-0.0901^{*}}$	−0.0507	$\boxed{-0.1211^{**}}$	−0.1191	$\boxed{-0.1414^{**}}$	$\boxed{-0.1568^{**}}$	$\boxed{-0.0970^{*}}$	$\boxed{-0.1330^{**}}$
	$\boxed{(0.0384)}$	$\boxed{(0.0478)}$	$\boxed{(0.0470)}$	$\boxed{(0.0518)}$	(0.0406)	$\boxed{(0.0606)}$	(0.0760)	$\boxed{(0.0668)}$	$\boxed{(0.0656)}$	$\boxed{(0.0527)}$	$\boxed{(0.0636)}$
Year controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of CEO transitions	4,692	2,397	3,239	1,533	2,022	2,312	2,012	2,045	1,882	2,174	2,015

even favorable). Once CEO choice is isolated using quasi-random variation, the performance cost of family CEOs is much larger.

4.7 Table VII: alternative windows and individual-level subsamples

What it reports. IV-2SLS estimates under different event windows and subsamples.

Key diagnostics.

- **Placebo windows:** pre-transition and long post-transition windows show no clear negative effect, supporting the interpretation that the effect is concentrated around the transition.
- **Departing CEO age restrictions:** estimates remain negative for typical retirement ages (55–70) and other ages.
- **Alternative instrument (CEO death):** using death around transition as an instrument also yields a negative estimate, supporting robustness to instrument choice.

4.8 Table VIII: heterogeneity by firm and industry characteristics

What it reports. IV estimates in subsamples by size, governance features, and industry environments (growth, R&D, wages, schooling, import penetration, volatility).

Main message. Negative effects appear broadly across many environments and are often statistically significant. There is limited evidence that "good" governance or "modern" industries fully offset the underperformance of family CEOs.

4.9 Table IX: alternative dependent variables and bankruptcy risk

Alternative profitability measures. Family CEOs reduce net income/assets and reduce return on capital employed, consistent with operating-profitability results.

Scale effects. The effect on log assets is imprecise, suggesting that the main action is on profitability rather than mechanical size changes.

Downside tail risk. Family CEOs increase the probability of bankruptcy/liquidation in the three years after transition, especially among firms in the bottom half of pre-transition profitability (column (5)).

TABLE IX
ALTERNATIVE DEPENDENT VARIABLES

	Net income to assets (1)	Return on capital employed (2)	Log of assets (3)	Bankruptcy/ liquidation (4)	Bankruptcy/ liquidation (5)
Family CEO	$\boxed{-0.0688^{**}}$	$\boxed{-0.1172^{**}}$	−0.3059	0.0588	0.1715*
	$\boxed{(0.0345)}$	$\boxed{(0.0455)}$	(0.2387)	(0.0553)	(0.1020)
Year controls	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes
Number of CEO transitions	4,692	2,553	4,692	4,568	2,258

Interpretation. Family CEO underperformance is economically meaningful: it shows up not only in average profitability but also in failure risk among vulnerable firms.

4.10 Table X: CCEO Characteristics by Type of Transition

The table compares *observable incoming CEO characteristics* across: (i) the instrument groups (departing CEO's firstborn child is male vs. female), (ii) *family* vs. *unrelated* CEO successions, and (iii) within *family* transitions only (firstborn-male family CEO vs. other family CEO). Incoming CEO characteristics include: prior CEO experience, prior outside board experience, years of schooling, and college/graduate attendance.

Panel A: By gender of the first child (reduced form). Male-firstborn firms (the group more likely to appoint a family CEO) select CEOs with slightly weaker observable credentials:

- Prior CEO experience: 0.2278 (male-firstborn) vs. 0.2509 (female-firstborn), difference −0.023 ($p < 0.10$).
- Prior outside board position: essentially no difference (−0.005, n.s.).
- Years of schooling: 13.2482 vs. 13.4476, difference −0.199 ($p < 0.05$).
- College/graduate attendance: 0.3809 vs. 0.4115, difference −0.031 ($p < 0.05$).

Interpretation: the instrument shifts firms toward CEO choices with lower measured education and (marginally) lower prior CEO experience, consistent with the instrument operating through a higher propensity for family succession.

Panel B: By family links (Family vs Unrelated) and IV-2SLS. Unrelated CEOs look substantially more "professional" than family CEOs in both experience and education:

- Prior CEO experience: family 0.1543 vs. unrelated 0.2853, gap −0.131 ($p < 0.01$).
- Prior board experience: family 0.2410 vs. unrelated 0.3125, gap −0.072 ($p < 0.01$).
- Schooling: family 12.659 vs. unrelated 13.677, gap −1.018 ($p < 0.01$).
- College/grad: family 0.2787 vs. unrelated 0.4511, gap −0.172 ($p < 0.01$).

Column (V) reports the **IV-2SLS** (LATE) difference in CEO traits induced by the instrument (firstborn male):

- Prior CEO experience: IV gap -0.242 ($p < 0.10$).
- Prior board experience: IV gap -0.048 (n.s.).
- Schooling: IV gap -1.992 ($p < 0.05$).
- College/grad: IV gap -0.306 ($p < 0.05$).

Interpretation: the "marginal" family CEOs identified by the IV do *not* appear observably better than the average family CEO; if anything, the IV gaps are similar or larger in magnitude. This supports the view that the IV estimates are not driven by selecting an unusually high-quality subset of family CEOs.

TABLE X
CEO CHARACTERISTICS BY TYPE OF TRANSITION

A. By gender of the first child:	(1) All	(2) First-child male	(3) First-child female	(4) Difference	(5) IV-2SLS
CEO previously held a CEO position	0.2417	0.2278	0.2509	−0.023*	
	(0.0059)	(0.0084)	(0.0092)	(0.0125)	
CEO previously held a board position	0.2887	0.2892	0.2938	−0.005	
	(0.0062)	(0.0091)	(0.0097)	(0.0133)	
Number of years of schooling by CEO	13.333	13.2482	13.4476	−0.199**	
	(0.0388)	(0.0561)	(0.0601)	(0.0822)	
CEO attended college or a graduate program	0.3928	0.3809	0.4115	−0.031**	
	(0.0068)	(0.0099)	(0.0106)	(0.0145)	

B. By family links:	Family	Unrelated	Difference	IV-2SLS
CEO previously held a CEO position	0.1543	0.2853	−0.131***	−0.242*
	(0.0086)	(0.0076)	(0.0114)	(0.130)
CEO previously held a board position	0.2410	0.3125	−0.072***	−0.048
	(0.0102)	(0.0078)	(0.0128)	(0.139)
Number of years of schooling by CEO	12.659	13.677	−1.018***	−1.992**
	(0.0595)	(0.0491)	(0.0771)	(0.818)
CEO attended college or a graduate program	0.2787	0.4511	−0.172***	−0.306**
	(0.0107)	(0.0085)	(0.0137)	(0.144)

TABLE X
(CONTINUED)

C. Family transitions only:	Firstborn male CEO	Other family CEO	Difference
CEO previously held a CEO position	0.1407	0.1583	0.018
	(0.0173)	(0.0099)	(0.0199)
CEO previously held a board position	0.2247	0.2458	0.021
	(0.0208)	(0.0116)	(0.0238)
Number of years of schooling by CEO	12.774	12.625	−0.149
	(0.1092)	(0.0700)	(0.1297)
CEO attended college or a graduate program	0.2730	0.2804	0.007
	(0.0222)	(0.0123)	(0.0254)

Panel C: Family transitions only (primogeniture check). Within the subsample of family transitions, firstborn-male family CEOs are not observably different from other family CEOs:

- Prior CEO experience: 0.1407 vs. 0.1583 (diff. 0.018, n.s.).
- Prior board experience: 0.2247 vs. 0.2458 (diff. 0.021, n.s.).
- Schooling: 12.774 vs. 12.625 (diff. -0.149, n.s.).
- College/grad: 0.2730 vs. 0.2804 (diff. 0.007, n.s.).

Interpretation: primogeniture-induced family CEOs do not look systematically different on observables, mitigating the concern that the instrument identifies a "particularly low-quality" subset of heirs.

Bottom line. Table X supports two conclusions: (i) unrelated CEOs are substantially more qualified on observable metrics than family CEOs, and (ii) the IV identifies marginal family CEO appointments that are not observably different from typical family CEOs—supporting the causal interpretation that family succession itself drives the performance differences.

5 Overall interpretation and economic meaning

5.1 Why IV estimates exceed OLS

An explanation is **selection on expected performance**: families may appoint family CEOs when prospects are strong or when the firm is less complex, which biases OLS toward finding small (or even positive) effects. The IV strategy isolates CEO choices shifted by firstborn gender and therefore reveals a larger negative causal effect.

5.2 Managerial-capital misallocation perspective

The paper can be read as a micro-foundation for a "managerial misallocation" wedge: if family firms systematically allocate top jobs to less qualified heirs, aggregate productivity can fall even without capital/labor distortions. This links naturally to empirical IO/productivity frameworks where misallocation of key inputs (including management) lowers measured TFP.

6 Conclusion

6.1 Summary

This paper concludes that **appointing a family CEO causally reduces firm performance around CEO successions**. The key challenge is endogeneity: families may choose family successors in systematically different circumstances than unrelated successors. The paper overcomes this by using a quasi-experimental instrument: **the gender of the departing CEO's firstborn child**. A male firstborn increases the probability that the successor is a family member (consistent with a primogeniture-like preference), but is plausibly unrelated to firm performance changes at the time of succession except through its impact on the succession decision.

Using this design, the paper delivers a chain of evidence:

(i) **Relevance:** a male firstborn sharply increases the likelihood of a family CEO (strong first stage).

(ii) **Reduced form:** a male firstborn predicts worse changes in profitability around succession.

(iii) **Causal effect (2SLS):** instrumenting family CEO status yields a large negative effect on profitability changes around succession.

(iv) **Selection explains IV > OLS in magnitude:** families tend to appoint family CEOs when observed and unobserved conditions are relatively favorable, which makes naive OLS comparisons understate the true performance cost.

(v) **Mechanism: managerial talent misallocation:** family CEOs are measurably less qualified (experience and education), consistent with family control considerations dominating merit in CEO assignment.

(vi) **Downside risk:** family CEOs increase failure risk (bankruptcy/liquidation) among already weak firms.

6.2 Key quantitative conclusions

The headline magnitudes can be summarized directly from the main tables:

(1) **First stage (Table V, Part A):**

$$\Pr(famCEO = 1 \mid \text{firstborn male} = 1) - \Pr(famCEO = 1 \mid \text{firstborn male} = 0) \approx 0.0955. \tag{5}$$

That is, a male firstborn increases the probability of appointing a family CEO by about **9.6 percentage points** (with large first-stage F-statistics, indicating a strong instrument).

(2) **Reduced form (Table V, Part B):**

$$\mathbb{E}[\Delta Perf \mid \text{firstborn male} = 1] - \mathbb{E}[\Delta Perf \mid \text{firstborn male} = 0] \approx -0.012. \tag{6}$$

A male firstborn predicts about a **1.2 percentage point** worse change in operating profitability around succession (for the industry-adjusted OROA change outcome).

(3) **OLS vs IV (Table VI):**

- OLS implies a modest underperformance: about -0.014 without year controls and -0.008 with controls.
- IV-2SLS implies a much larger causal effect: about -0.126 in the baseline IV specification.
- Using alternative male-child instruments yields effects in the same direction, typically in the -0.06 **to** -0.09 range.
- Using the industry-and-performance-adjusted profitability measure, the preferred IV estimate is about -0.0886.

(4) **CEO quality differences (Table X, Part B):** Relative to unrelated CEOs, family CEOs are less qualified on observables:

- prior CEO experience is lower (difference ≈ -0.131; IV gap ≈ -0.242),
- schooling is lower (difference ≈ -1.018 years; IV gap ≈ -1.992 years),
- college attendance is lower (difference ≈ -0.172; IV gap ≈ -0.306).

(5) **Downside tail risk (Table IX):** The probability of bankruptcy/liquidation within three years post-succession rises by about **0.17** among firms in the bottom half of pre-succession profitability (statistically meaningful at conventional levels).

6.3 Economic intuition: why family CEOs reduce performance

The results are naturally understood as **managerial talent misallocation driven by private benefits of control**.

1) A simple objective that rationalizes family succession. Suppose the controlling family chooses between an unrelated CEO (U) and a family CEO (F). Let firm profit under CEO type $s \in \{F, U\}$ be π_s, and let the family obtain private benefits B_s (control, trust, dynastic value, private consumption) that are larger under a family CEO:

$$\max_{s \in \{F,U\}} \quad \pi_s + B_s, \qquad \text{with } B_F > B_U. \tag{7}$$

Even if $\pi_F < \pi_U$ (family CEO is less competent on average), the family may still choose F when $B_F - B_U$ is sufficiently large. This generates a systematic wedge between *family-optimal* and *firm-value-optimal* CEO assignment.

2) Why the firstborn-male instrument matters. The gender of the firstborn child shifts the probability of selecting F because it affects the availability and perceived suitability of a family successor (consistent with male primogeniture norms), but it is plausibly orthogonal to contemporaneous firm shocks at the time of succession. Thus the instrument supplies quasi-random variation in whether a firm appoints a family CEO.

3) Why IV effects are much larger than OLS. OLS compares firms that *choose* family CEOs to those that do not. If families tend to appoint heirs when conditions look good (or when the firm is easier to run), then the OLS comparison is biased toward finding small underperformance. IV isolates CEO appointments shifted by quasi-random variation in successor gender, revealing a larger negative causal effect.

4) Why the mechanism is consistent with observed CEO characteristics. If family succession is driven by dynastic/control motives, one should observe weaker average observable qualifications among family CEOs. Table X matches this prediction: family CEOs have less prior CEO experience and lower educational attainment on average, and the marginal family CEOs induced by the instrument are not observably better than typical family CEOs.

6.4 Interpretation of what the IV identifies

The 2SLS estimate should be read as a **local average treatment effect**: it captures the performance impact of appointing a family CEO for **compliers**—firms whose family-CEO decision is shifted by having a male vs female firstborn child. In practice, these are firms on the margin between appointing a family or an unrelated CEO, which is exactly where policy-relevant distortions in CEO assignment are most likely to be identified.

6.5 Takeaway

Using the departing CEO's firstborn gender as an instrument, the paper shows that family CEOs causally reduce operating profitability around CEO successions—with IV effects far larger than OLS due to selection—and the evidence is consistent with dynastic succession generating managerial talent misallocation and higher failure risk among weak firms.